

## SAFETY DATA SHEET

Creation Date 06-Jul-2010

Revision Date 25-Mar-2024

Revision Number 2

### 1. Identification

**Product Name** Hydrofluoric acid, 40% aq.  
**Cat No. :** R21727  
**Synonyms** Hydrofluoric acid solution; Fluohydric acid; Fluoric acid  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |              |
|--------------------------------------------------|--------------|
| Corrosive to metals                              | Category 1   |
| Acute oral toxicity                              | Category 2   |
| Acute dermal toxicity                            | Category 1   |
| Acute Inhalation Toxicity - Vapors               | Category 2   |
| Skin Corrosion/Irritation                        | Category 1 A |
| Serious Eye Damage/Eye Irritation                | Category 1   |
| Specific target organ toxicity (single exposure) | Category 3   |
| Target Organs - Respiratory system.              |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals

Causes severe skin burns and eye damage  
May cause respiratory irritation  
Fatal if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Do not get in eyes, on skin, or on clothing  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear respiratory protection  
Keep only in original container

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Spills**

Absorb spillage to prevent material damage

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in corrosive resistant polypropylene container with a resistant inliner  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Water             | 7732-18-5 | 60       |
| Hydrogen fluoride | 7664-39-3 | 40       |

### 4. First-aid measures

**General Advice**

Immediate and specialised first aid and medical treatment is required. Speed is of the essence. Flush with plenty of water immediately. Continue flushing during transport to hospital or medical center.

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required. Dermal burns may be treated with calcium gluconate gel or slurry in water or glycerine. This compound binds the active fluorides in an insoluble form and limits burn extension and pain. Soaking or immersion with iced 0.13% Benzalkonium chloride solution may be used for skin burns and should be continued until the pain is relieved. Do not use in eyes. |
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. A nebulized solution of 2.5% Calcium gluconate may be administered with Oxygen by inhalation.                                                 |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                                                                                                                                                              |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | No information available                                                 |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                 |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Gaseous hydrogen fluoride (HF).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

Health  
4

Flammability  
0

Instability  
1

Physical hazards  
N/A

## 6. Accidental release measures

|                             |                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b> | Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Environmental Precautions** Should not be released into the environment.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store in metal containers. Incompatible Materials. Metals. Cyanides. Sulfides. Bases. Fluorine.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | ACGIH TLV                                                         | OSHA PEL                                                                                            | NIOSH                                                                                                                                  | Mexico OEL (TWA)                                          |
|-------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Hydrogen fluoride | TWA: 0.5 ppm TWA: 2.5 mg/m <sup>3</sup><br>Ceiling: 2 ppm<br>Skin | (Vacated) TWA: 3 ppm<br>(Vacated) TWA: 2.5 mg/m <sup>3</sup><br>(Vacated) STEL: 6 ppm<br>TWA: 3 ppm | IDLH: 30 ppm IDLH: 250 mg/m <sup>3</sup><br>TWA: 3 ppm<br>TWA: 2.5 mg/m <sup>3</sup><br>Ceiling: 6 ppm<br>Ceiling: 5 mg/m <sup>3</sup> | TWA: 0.5 ppm TWA: 2.5 mg/m <sup>3</sup><br>Ceiling: 2 ppm |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

**Eye/face Protection** Tight sealing safety goggles. Face protection shield.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Acid gases filter. Type E. Yellow. conforming to EN14387. Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                     |                          |
|---------------------|--------------------------|
| Physical State      | Liquid                   |
| Appearance          | Colorless                |
| Odor                | pungent                  |
| Odor Threshold      | No information available |
| pH                  | < 1.0                    |
| Melting Point/Range | -35 °C / -31 °F          |
| Boiling Point/Range | 105 °C / 221 °F          |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | 2.21                     |
| Specific Gravity                       | 1.15-1.20                |
| Solubility                             | miscible                 |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | H F                      |
| Molecular Weight                       | 20                       |

## 10. Stability and reactivity

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                  |
| Stability                        | Stable under normal conditions.                                             |
| Conditions to Avoid              | Incompatible products. Excess heat.                                         |
| Incompatible Materials           | Metals, Cyanides, Sulfides, Bases, Fluorine                                 |
| Hazardous Decomposition Products | Gaseous hydrogen fluoride (HF)                                              |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                    |
| Hazardous Reactions              | Corrosive to metals. Contact with metals may evolve flammable hydrogen gas. |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

Oral LD50 Category 2. ATE = 5 - 50 mg/kg.

Dermal LD50 Category 1. ATE < 50 mg/kg.

Vapor LC50 Category 2. ATE = 0.5 - 2 mg/l.

#### Component Information

| Component         | LD50 Oral  | LD50 Dermal | LC50 Inhalation              |
|-------------------|------------|-------------|------------------------------|
| Water             | -          | -           | -                            |
| Hydrogen fluoride | Not listed | Not listed  | LC50 = 0.79 mg/L ( Rat ) 1 h |

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes severe burns by all exposure routes                                               |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component         | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|-----------|------------|------------|------------|------------|------------|
| Water             | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Hydrogen fluoride | 7664-39-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                  |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                  |
| <b>STOT - single exposure</b>                     | Respiratory system                                                                                                                                                                                                                                         |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

| Component         | Freshwater Algae | Freshwater Fish                          | Microtox   | Water Flea                                |
|-------------------|------------------|------------------------------------------|------------|-------------------------------------------|
| Hydrogen fluoride | Not listed       | LC50 = 660 mg/L, 48h<br>(Leuciscus idus) | Not listed | EC50 = 270 mg/L, 48h<br>(Daphnia species) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available. Miscible with water

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component         | log Pow |
|-------------------|---------|
| Hydrogen fluoride | -1.4    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                     | RCRA - U Series Wastes | RCRA - P Series Wastes |
|-------------------------------|------------------------|------------------------|
| Hydrogen fluoride - 7664-39-3 | U134                   | -                      |

## 14. Transport information

### DOT

|                                |                   |
|--------------------------------|-------------------|
| <b>UN-No</b>                   | UN1790            |
| <b>Proper Shipping Name</b>    | HYDROFLUORIC ACID |
| <b>Hazard Class</b>            | 8                 |
| <b>Subsidiary Hazard Class</b> | 6.1               |
| <b>Packing Group</b>           | II                |

### TDG

|                                |                   |
|--------------------------------|-------------------|
| <b>UN-No</b>                   | UN1790            |
| <b>Proper Shipping Name</b>    | HYDROFLUORIC ACID |
| <b>Hazard Class</b>            | 8                 |
| <b>Subsidiary Hazard Class</b> | 6.1               |
| <b>Packing Group</b>           | II                |

### IATA

|              |        |
|--------------|--------|
| <b>UN-No</b> | UN1790 |
|--------------|--------|

**Proper Shipping Name** HYDROFLUORIC ACID  
**Hazard Class** 8  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

**IMDG/IMO**

**UN-No** UN1790  
**Proper Shipping Name** HYDROFLUORIC ACID  
**Hazard Class** 8  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

## 15. Regulatory information

**United States of America Inventory**

| Component         | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Water             | 7732-18-5 | X    | ACTIVE                                        | -                           |
| Hydrogen fluoride | 7664-39-3 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

TSCA 12(b) - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component         | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Water             | 7732-18-5 | X   | -    | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Hydrogen fluoride | 7664-39-3 | X   | -    | 231-634-8 | X     | X    | X    | X    | X     | KE-20198 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component         | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-------------------|-----------|----------|-------------------------------|---------------------------------|
| Hydrogen fluoride | 7664-39-3 | 40       | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component         | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Hydrogen fluoride | X                          | 100 lb                      | -                      | -                         |

**Clean Air Act**

| Component | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-----------|-----------|-------------------------|-------------------------|
|-----------|-----------|-------------------------|-------------------------|

|                   |   |  |   |
|-------------------|---|--|---|
| Hydrogen fluoride | X |  | - |
|-------------------|---|--|---|

OSHA - Occupational Safety and Health Administration Not applicable

| Component         | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-------------------|----------------------------------|----------------------------|
| Hydrogen fluoride | -                                | TQ: 1000 lb                |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component         | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|-------------------|--------------------------|-------------------------------------------|-------------------------------|
| Hydrogen fluoride | 100 lb                   | 100 lb                                    | 100 lb<br>45.4 kg             |

California Proposition 65 This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Water             | -             | -          | X            | -        | -            |
| Hydrogen fluoride | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product contains the following DHS chemicals:  
**Legend** - STQs = Screening Threshold Quantities, APA = A placarded amount

| Component         | DHS Chemical Facility Anti-Terrorism Standard                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Hydrogen fluoride | Release STQs - 1000lb (concentration >=50%)<br>Release STQs - 1000lb (anhydrous)<br>Theft STQs - 45lb (anhydrous) |

**Other International Regulations**

Mexico - Grade No information available

**Authorisation/Restrictions according to EU REACH**

| Component         | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water             | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Hydrogen fluoride | 7664-39-3 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>



## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component         | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water             | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Hydrogen fluoride | 7664-39-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component         | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water             | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Hydrogen fluoride | 7664-39-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y34                      |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

06-Jul-2010

## Revision Date

25-Mar-2024

## Print Date

25-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**